

```
In [1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [2]: iris = sns.load_dataset('iris')
```

```
In [3]: iris
```

```
Out[3]:
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

150 rows × 5 columns

```
In [6]: iris.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   sepal_length    150 non-null   float64
1   sepal_width     150 non-null   float64
2   petal_length    150 non-null   float64
3   petal_width     150 non-null   float64
4   species         150 non-null   object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
In [7]: iris.describe()
```

Out[7]:

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

In [9]: `iris.dtypes`

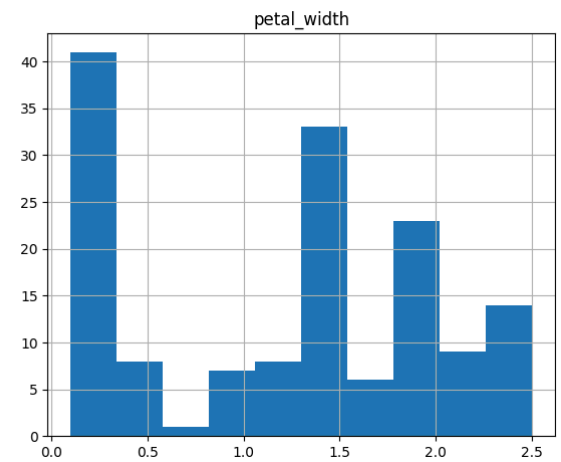
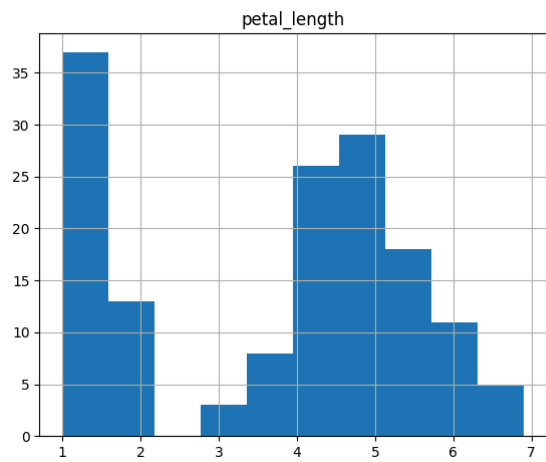
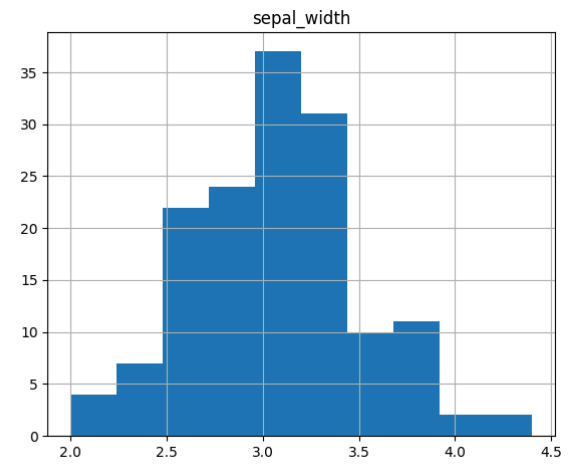
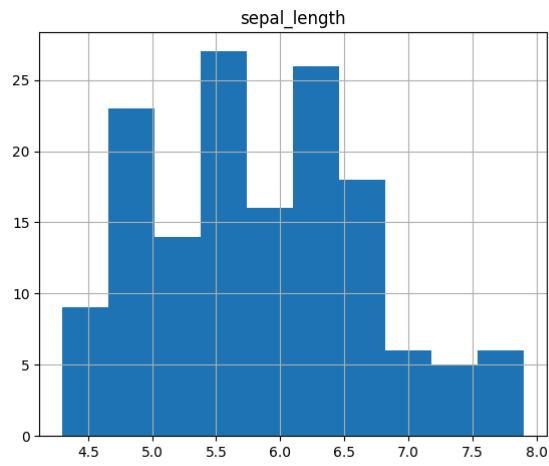
Out[9]:

sepal_length	float64
sepal_width	float64
petal_length	float64
petal_width	float64
species	object
dtype:	object

In [13]: `iris.hist(figsize = (15,12))`

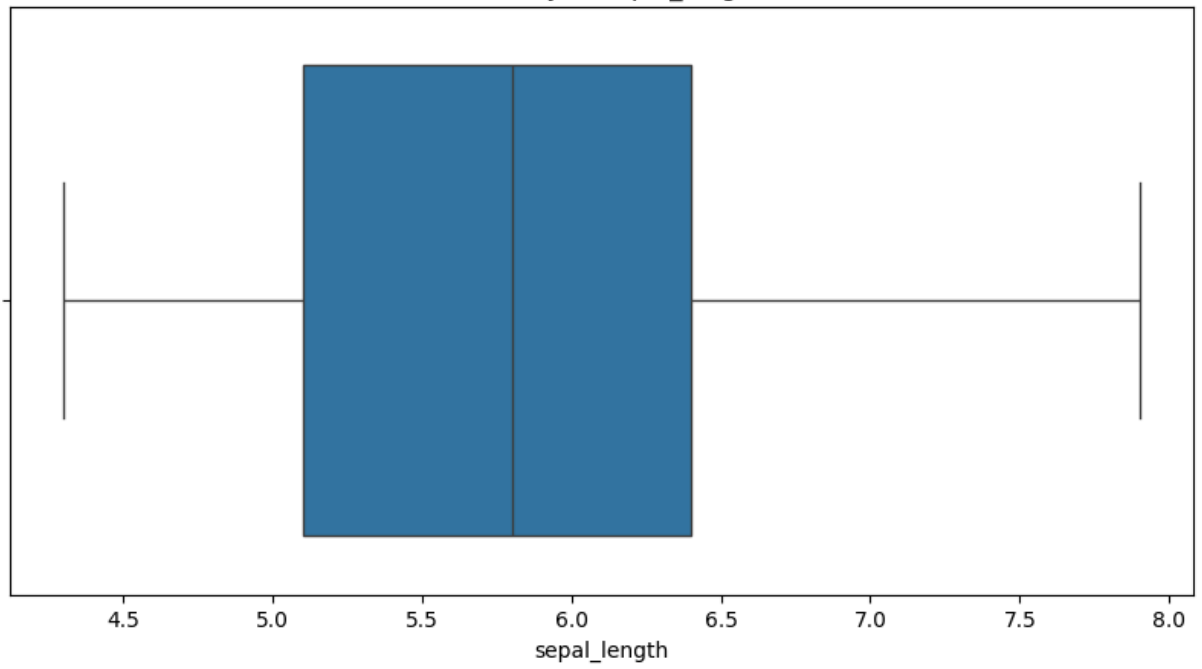
Out[13]:

```
array([[<Axes: title={'center': 'sepal_length'}>,
        <Axes: title={'center': 'sepal_width'}>],
       [<Axes: title={'center': 'petal_length'}>,
        <Axes: title={'center': 'petal_width'}>]], dtype=object)
```

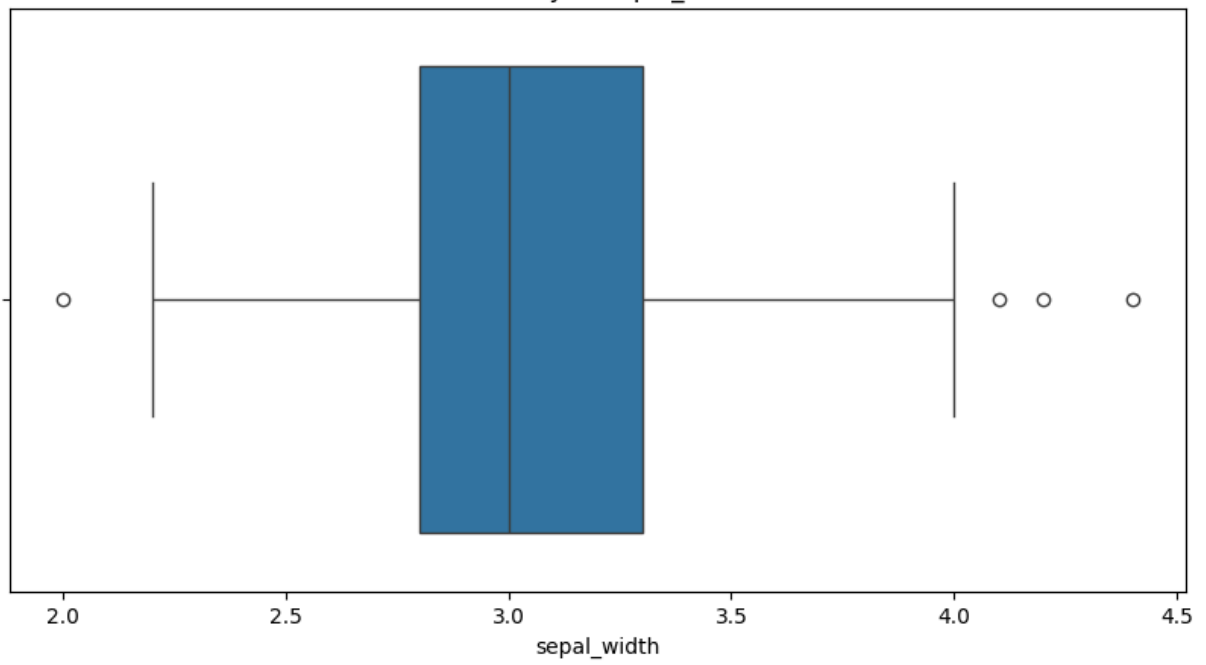


```
In [25]: for col in iris.columns[:-1]:  
          plt.figure(figsize = (10,5))  
          sns.boxplot(x=col,data = iris)  
          plt.title(f"Summary of {col}")  
          plt.show()
```

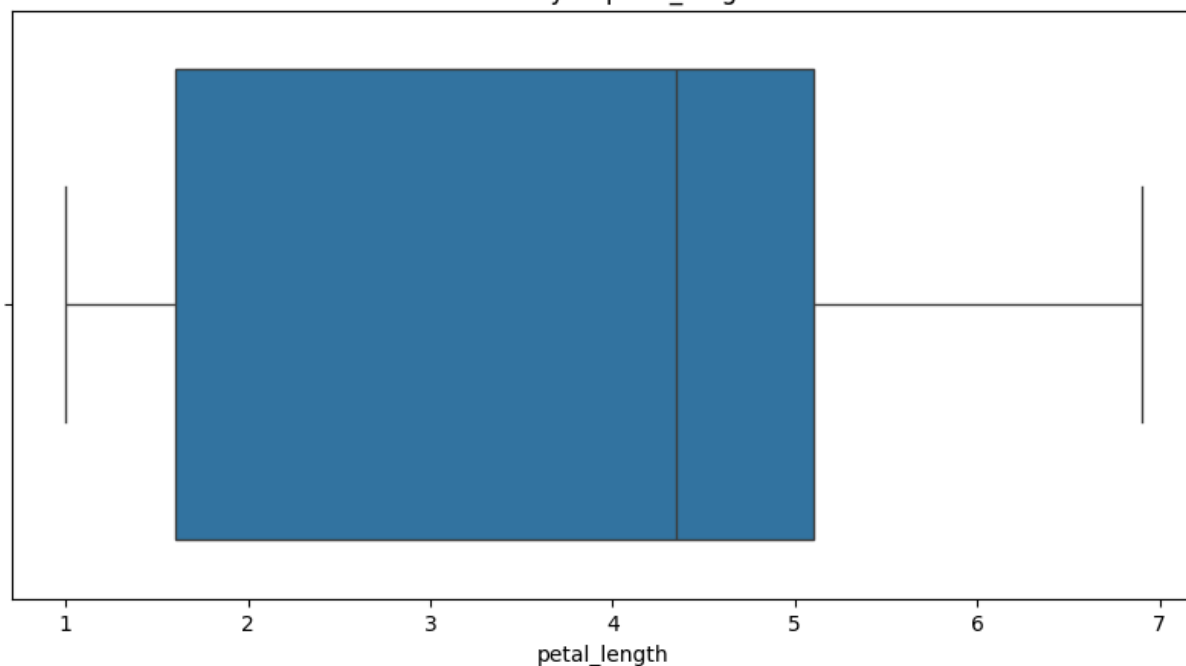
Summary of sepal_length



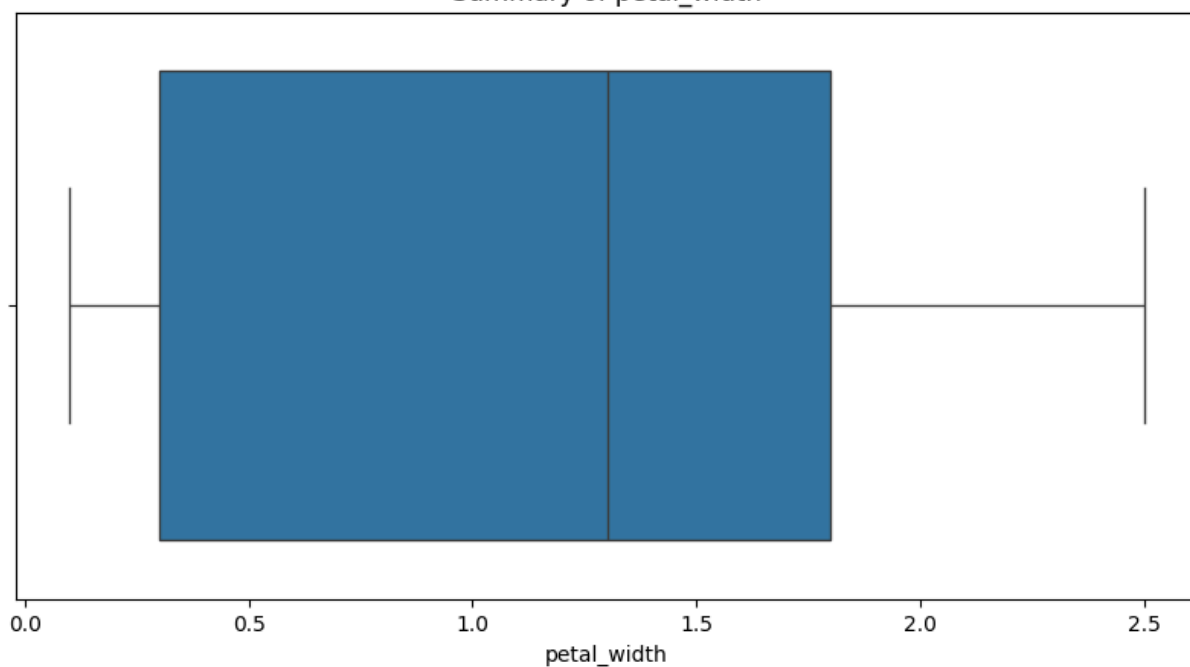
Summary of sepal_width



Summary of petal_length



Summary of petal_width



```
In [50]: for column in iris.columns[:-1]: # Exclude 'species' column
    Q1 = iris[column].quantile(0.25)
    Q3 = iris[column].quantile(0.75)

    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = iris[column].apply(lambda x : x if ( x < lower_bound or x > upper_bound )
    # outliers = outliers.dropna())
```

```

print(f"\nFeature: {column}")
print(f"  Mean: {iris[column].mean():.2f}, Median: {iris[column].median():.2f},
print(f"  Outliers Detected: {'No' if outliers.empty else 'Yes'}", "\n  " f"Outl
if not outliers.empty:
    sns.boxplot(x = column, data = iris)
print("-" * 40)

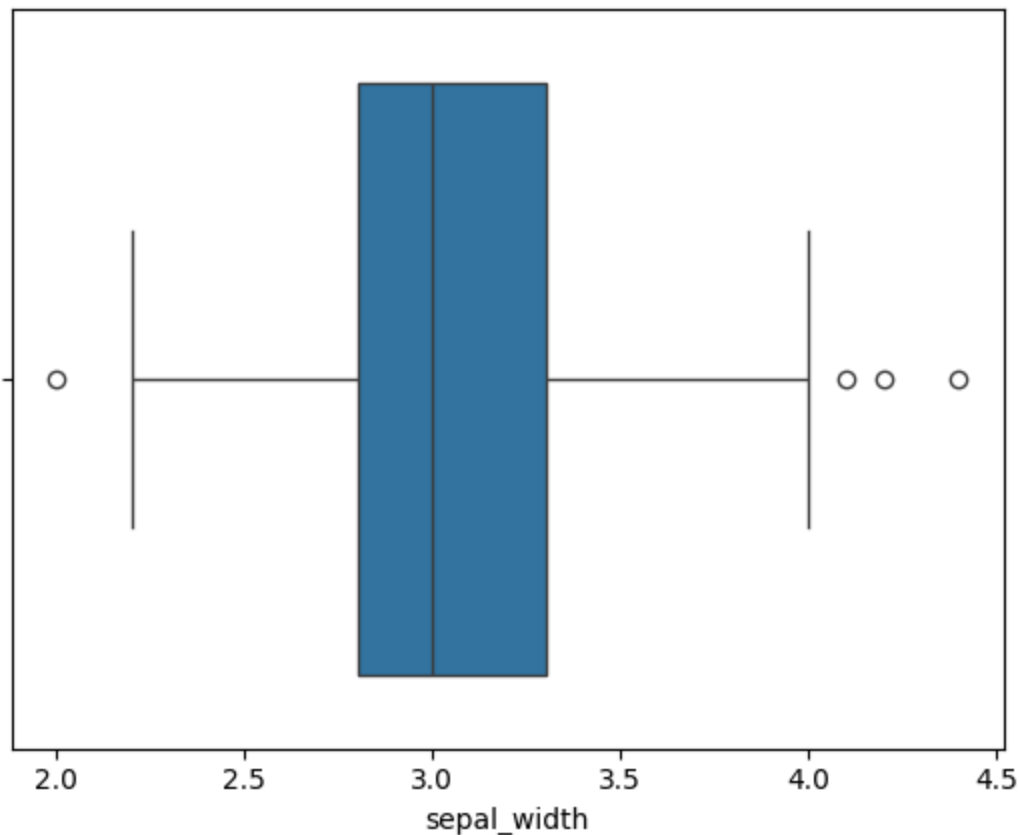
```

Feature: sepal_length
 Mean: 5.84, Median: 5.80, Std Dev: 0.83
 Outliers Detected: No

Feature: sepal_width
 Mean: 3.06, Median: 3.00, Std Dev: 0.44
 Outliers Detected: Yes
 Outlier Values: [4.4, 4.1, 4.2, 2.0]

Feature: petal_length
 Mean: 3.76, Median: 4.35, Std Dev: 1.77
 Outliers Detected: No

Feature: petal_width
 Mean: 1.20, Median: 1.30, Std Dev: 0.76
 Outliers Detected: No



In []: